

Final Project Summary Report: Fake News Classification using LSTM

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I. INTRODUCTION

A. Project Background

Fake news is false or misleading information presented as news. In the modern digital age, where there are countless venues for exchanging data, the broad issue of fake news is exceedingly challenging to address. The proliferation of deluding data such as fake news and phony audits on news web journals, online publications, and internet business apps has been aided by the availability of the web, cell phones, and social media. Individuals can quickly fabricate comments and news on social media. The most difficult challenge nowadays is determining which news is real or fake.

B. Project Goal

Therefore, I intend to use some models such as LSTM to classify a dataset from Kaggle containing true and false news, which will be divided into training and test data, and compare their performance on the same dataset. I hope that through this project I can improve my understanding between different models and hone my skills in actually pre-processing data as well as analyzing it.

C. Literature Review

There have been many research efforts in the field of fake news classification. These efforts range from manual fact-checking to developing advanced Machine Learning algorithms. From a comprehensive review of several literature[1][2], which compare different machine learning approaches in terms of performance on different datasets, such as traditional learning models, deep learning models, and advanced pre-trained language models like BERT, we can see that ML algorithms perform better on short text classification when the text is one or two lines, which is also efficient in computation. In contrast, transformers outperform the other algorithms in working on longer text. More importantly, Ref[1] found that pre-trained BERT-based models are robust to the size of the dataset and can perform significantly better on very small sample size. They also found that Naive Bayes with n-gram can attain similar results to neural network-based models on a dataset when the dataset size is sufficient. The performance of LSTM-based models greatly depends on the length of the dataset as well as the information given in a news article. With adequate information provided in a news article, LSTM-based models have a higher probability of overcoming

overfitting. Moreover, Ref[3] introduced adversarial training as a generalization to enhance the robustness of their model, and significantly improved the performance of all models compared to baseline methods

II. METHOD

A. Problem Formulation

Fake news classification is a supervised learning problem aimed at determining whether a piece of news is real or fake based on textual content and/or associated metadata.

Input:

A text sample, typically a news article, headline, or paragraph.

Output:

A binary label y :

$y=1$: Fake news.

$y=0$: Real news.

Alternatively, a multi-class label (e.g., true, false, partially true).

B. Dataset Description

The total rows of the dataset is 26000, including columns 'id', 'title', 'author', 'text', and there are nearly 3000 NaN values in 'title' and 'author'. If the 'label' is 1, the news is fake, and vice versa. Before applying algorithms, the dataset is split with a ratio of 80% to 20%, respectively, for training and testing.

C. Evaluation Measures

We usually use evaluation measures below to assess the performance of models in fake news classification, which are the same as the literature used we refer to.

Precision refers to the ratio of true positive instances, i.e. the instances that are correctly predicted as positive, to all instances predicted as positive. This metric measures the accuracy of positive predictions made by the model.

$$Precision = \frac{TP}{TP + FP}$$

Recall measures the ratio of true positive instances, i.e. correctly predicted positive instances, to all true positive instances

present in the dataset. This metric evaluates the model's ability to identify positive instances from the dataset accurately.

$$Recall = \frac{TP}{TP + FN}$$

Accuracy refers to the proportion of correctly classified instances out of all the instances in the dataset. It is a measure of how well the model performs overall.

$$Accuracy = \frac{TP + TN}{TP + FN + TN + FP}$$

The F1-score is a metric that considers the model's precision and recall to evaluate its overall accuracy. It is calculated as the harmonic mean of precision and recall, providing a more balanced assessment of the model's performance than accuracy alone.

$$F1 - score = 2 \times \frac{Pr \times Re}{Pr + Re}$$

whereas TP stands for True Positive News—news that is real and anticipated by the model to be real. TN stands for True Negative News—real news predicted to be fake by the model. FP stands for False Positive News—the fake news that the model incorrectly anticipated to be true. FN stands for False Negative News—the actual fake news predicted to be fake by the model.

D. Model Formulation

I formulate the binary classification model by using Keras, setting an embedding layer, a single LSTM layer with 128 nodes, a dense layer and two dropout layers which randomly sets 20% of the input units to 0 during training in order to prevent overfitting. Moreover, I set the loss function "binary_crossentropy" and Adam optimizer with a learning rate of 0.001.

Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	(None, 20, 40)	200,000
dropout (Dropout)	(None, 20, 40)	0
lstm (LSTM)	(None, 128)	86,528
dropout_1 (Dropout)	(None, 128)	0
dense (Dense)	(None, 1)	129

Total params: 286,657 (1.09 MB)
Trainable params: 286,657 (1.09 MB)

Fig. 1. Summary of LSTM Model

III. RESULTS

As the TABLE I shows below, all metrics of the LSTM model are close to 1, indicating that the model performs very well in classifying fake and real news.

Moreover, I compared the accuracy of LSTM with other models (TABLE II) and found that although LSTM requires

TABLE I
CLASSIFICATION REPORT OF LSTM MODEL

	Precision	Recall	F1-score	Support
0	0.99	0.99	0.99	1039
1	0.99	0.99	0.99	1041
Accuracy			0.99	2080
Macro Avg	0.99	0.99	0.99	2080
Weighted Avg	0.99	0.99	0.99	2080

more computation time, its performance in terms of classification accuracy is significantly superior. This result highlights the core strength of the LSTM algorithm: its ability to effectively capture temporal features in input data. Unlike traditional models, LSTM uses its memory cells and gating mechanisms to avoid the vanishing gradient problem, ensuring stable performance. This capability makes LSTM particularly well-suited for tasks that require analyzing sequential data, such as natural language processing.

TABLE II
PREDICTION RESULTS OF DIFFERENT MODELS

Model	Accuracy
Logistic Regression	0.698077
Naive Bayes	0.680288
Decision Tree	0.894712
Random Forest	0.911538
XGBOOST	0.975000
CatBoost	0.974519
LSTM	0.989423

CONCLUSION

After comparing the LSTM model with various traditional machine learning models for the task of fake news classification, the results demonstrate the superior performance of the LSTM model in terms of accuracy, while the computational cost of LSTM is relatively high compared to other models. Although my limited familiarity with model implementation led me to use only a single-layer LSTM model, the results already demonstrate the potential of LSTM in processing textual data. I believe that in the future, by combining LSTM with other more advanced algorithms, it will exhibit even greater potential and achieve more efficient computation in tackling the problem of fake news detection.

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